

ASSIGNMENT # 1

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Section: BSCS-4K

QUESTION # 1

1) Width of Interval $\frac{X_m - X_L}{6} = \frac{150 - 88}{6} = \lceil 10.33 \rceil = 11$

CLASS INTERVAL	CLASS BOUND	Freq	C.Freq	$\bar{x}$	fx	
86 — 96	85.5 — 96.5	3	3	91	273	X
97 — 107	96.5 — 107.5	8	11	102	816	X
108 — 118	107.5 — 118.5	15	26	113	1695	of 1 of 2 → Median
119 — 129	118.5 — 129.5	11	37	124	1364	of 3
130 — 140	129.5 — 140.5	10	47	135	1350	
141 — 151	140.5 — 151.5	3	50	146	438	
		$\Sigma f = 50$			$\Sigma fx = 5936$	

(a)

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{5936}{50} = 118.72$$

$$\text{Median} = L + \frac{h}{f} \left(\frac{\Sigma f}{2} - c.f \right)$$

$$\therefore \frac{\Sigma f}{2} = 25$$

$$\text{so } L = 107.5, h = 11, f_{\text{req}} = 15, c.f = 11$$

$$\text{Median} = 107.5 + \frac{11}{15} (25 - 11)$$

$$\text{Median} = 107.5 + 30.27$$

$$\text{Median} = 137.77$$

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$$\text{Mode} = L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times h$$

$$L = 107.5, f_m = 15, f_1 = 8, f_2 = 11, h = 11$$

$$\text{Mode} = 107.5 + \left(\frac{15 - 8}{30 - 8 - 11} \right) \times 11$$

$$\text{Mode} = 107.5 + \left(\frac{7}{11} \right) \times 11$$

$$\text{Mode} = 114.5$$

(b)

$$\text{Range} = X_m - X_L = 62$$

Class Interval	freq	x	fx	x^2	fx^2
86 - 96	3	91	273	8281	24843
97 - 107	8	102	816	10404	83232
108 - 118	15	113	1695	12769	191535
119 - 129	11	124	1364	15376	169136
130 - 140	10	135	1350	18225	182250
141 - 151	3	146	438	21316	63948
	$\Sigma f = 50$		$\Sigma fx = 5936$		$\Sigma fx^2 = 714944$

$$S^2 = \frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2$$

$$S^2 = \frac{714944}{50} - (118.72)^2$$

$$S^2 = 14298.88 - 14094.43$$

$$S^2 = 204.45$$

$$S = 14.29$$

(c)

Intervals	From	To	Freq	Proportion
$\bar{y} \pm s$	104.43	133.01	31	62%
$\bar{y} \pm 2s$	90.14	147.3	46	92%
$\bar{y} \pm 3s$	75.85	161.59	50	100%

Upon comparing with empirical rule our data is slightly flatter in the middle than a perfect normal distribution, but it follows the general mound-shaped pattern reasonably well.

Outliers:

Since our 3s interval is 75.85 to 161.59 and all our data points lied within this range, so there are no outliers detected using this rule.

(d) $Q_{\min} = 88$

$Q_{\max} = 150$

$$Q_1 = L + \frac{h}{f} \left(i \frac{\sum f}{4} - c.f \right)$$

$$\therefore \frac{\sum f}{4} = 12.5$$

$$Q_1 = 107.5 + \frac{11}{15} (12.5 - 11)$$

$$Q_1 = 108.6$$

$$Q_2 = L + \frac{h}{f} \left(2 \frac{\sum f}{4} - c.f \right) = 107.5 + \frac{11}{15} (25 - 11)$$

$$Q_2 = 107.5 + \frac{11}{15} (25 - 11) = Q_2 = 117.77$$

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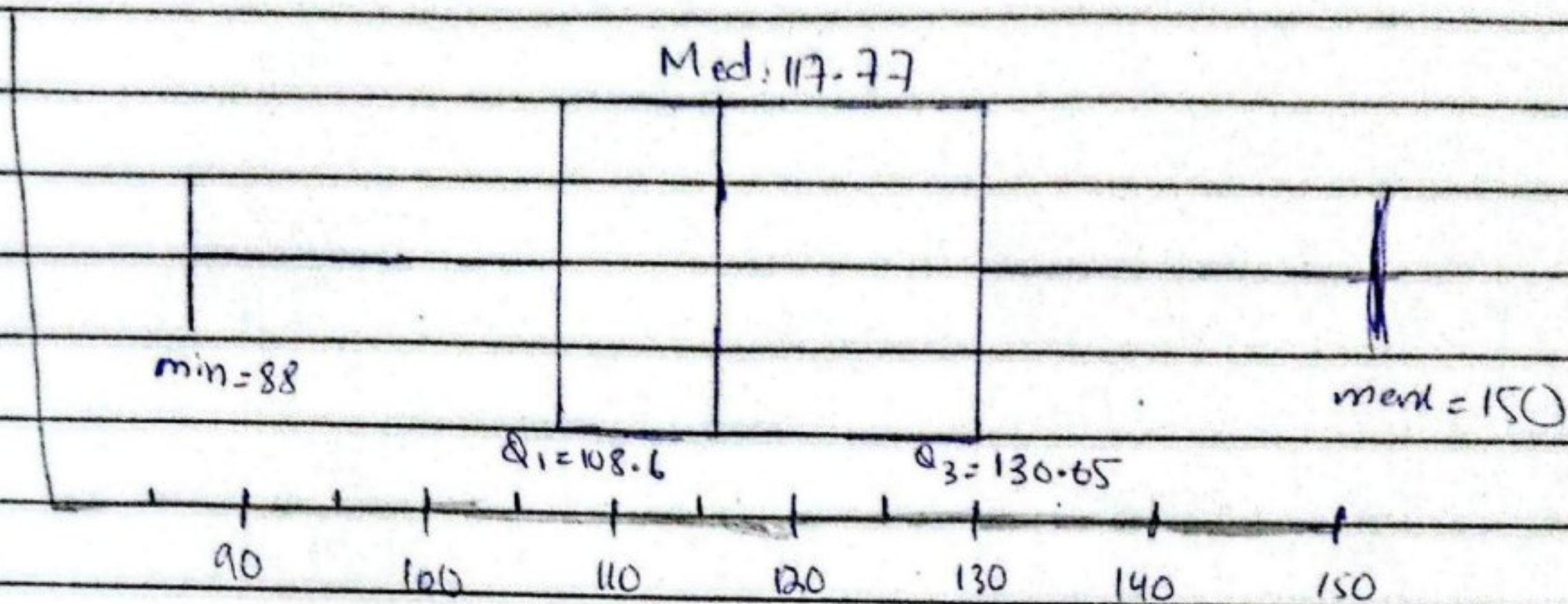
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$$Q_3 = L + \frac{h}{f} \left(\frac{3 \Sigma f}{4} - c.f \right)$$

$$Q_3 = 129.5 + \frac{11}{10} (37.5 - 37)$$

$$Q_3 = 130.05$$



$$IQR = Q_3 - Q_1 = 130.05 - 108.6 = 21.45$$

$$\text{Upper bound} = Q_3 + 1.5 \times IQR = 130.05 + (1.5 \times 21.45) = 162.225$$

$$\text{Lower bound} = Q_1 - 1.5 \times IQR = 108.6 - (1.5 \times 21.45) = 76.425$$

since all the values are in the interval so there are no outliers.

$$(e) P_{70} = L + \frac{h}{f} \left(\frac{70 \Sigma f}{100} - c.f \right) \because f = 35$$

$$P_{70} = 118.5 + \frac{11}{11} (35 - 26)$$

$$P_{70} = 118.5 + 9$$

$$P_{70} = 127.5$$

Question #2

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Row No	Marks	$x - \bar{x}$	$(x - \bar{x})^2$	Grade	Row No	Marks	$x - \bar{x}$	$(x - \bar{x})^2$	Grade
1	76	8.36	69.88	C	23	43	1	607.12	F
2	63	-4.64	21.52	D	24	78		107.32	C
3	57	-10.64	113.20	D	25	86		337.08	B
4	40		763.96	F	26	67		0.4096	D
5	42		657.4	F	27	55		159.76	D
6	65		6.96	D	28	88		414.52	B
7	72		19.00	C	29	60		58.36	D
8	73		28.72	C	30	82		206.2	C
9	95		743.56	B	31	96		804.28	B
10	80		152.76	C	32	68		0.1296	C
11	75		54.16	C	33	71		11.28	C
12	43		607.12	F	34	69		1.84	C
13	68		0.1296	C	35	27		1651.60	F
14	64		13.24	D	36	97		862.009	B
15	78		107.32	C	37	47		426	F
16	68		0.1296	C	38	83		285.42	C
17	68		159.76	D	39	74		40.44	C
18	75		54.16	C	40	63		21.52	D
19	85		286.88	E	41	77		87.60	C
20	77		87.60	C	42	77		87.60	C
21	40		763.96	F	43	82		206.20	C
22	79		129.04	C	44	50		311.16	E

(a) Descriptive Stats

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$$\text{Mean} = \frac{\sum \text{Marks}}{\sum \text{total sts}} = \frac{2976}{44} = 67.64$$

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n} = \frac{11474.18}{44} = 260.78$$

$$\text{Std dev} = \sqrt{260.78} = 16.15$$

$$\text{Median} = \frac{\left(\frac{n}{2} + \left(\frac{n}{2} + 1\right)\right)}{2} = \frac{22^{\text{th}} + 23^{\text{th}}}{2} = \frac{69 + 71}{2} = 70$$

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Mode = 66.77

(b)

A = Marks $> 4 + 26 = 99.94$ Marks ≥ 99.94

B = $83.79 \leq$ Marks < 99.94

C = $67.64 \leq$ Marks < 83.79

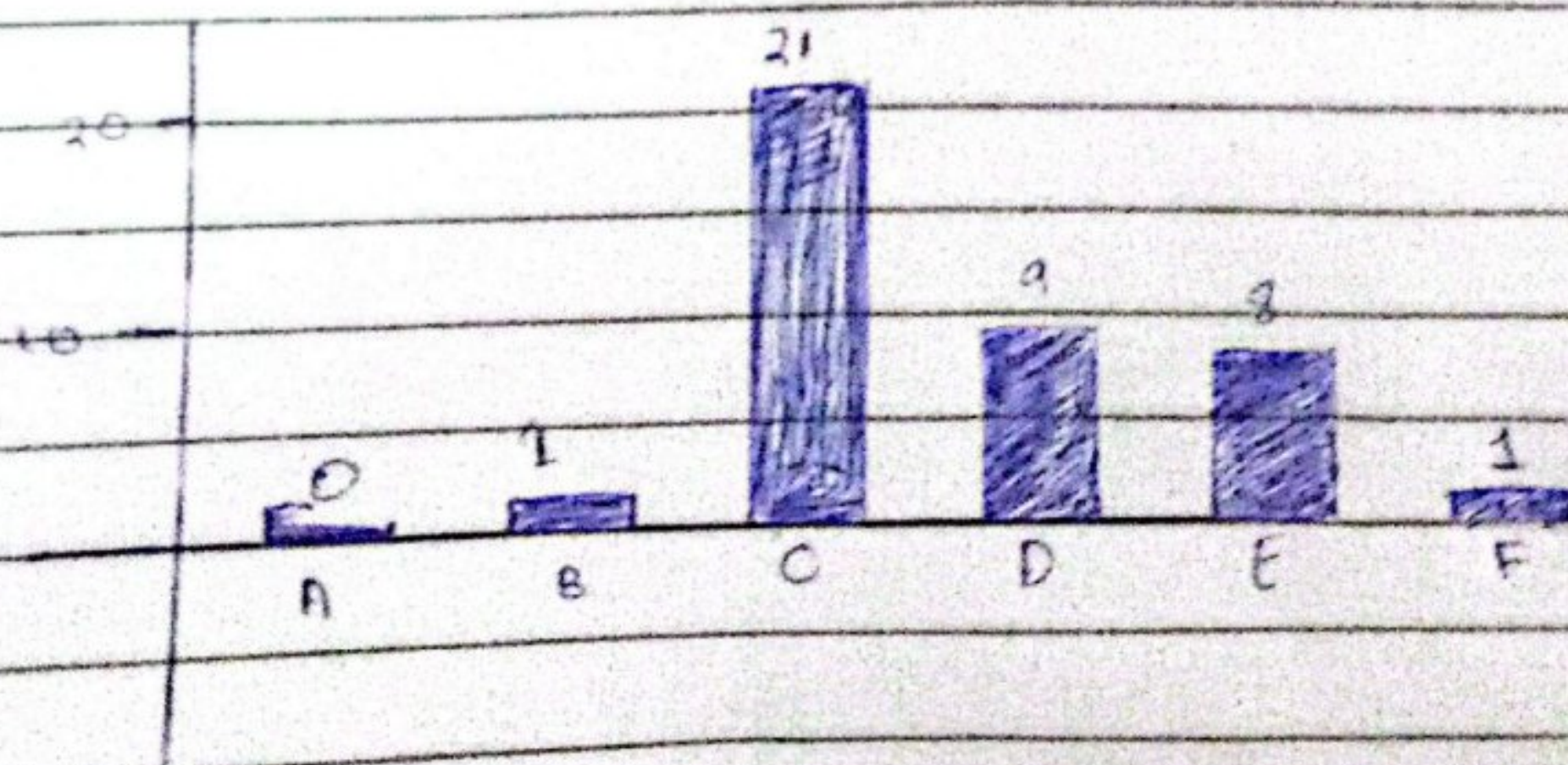
D = $51.49 \leq$ Marks < 67.64

E = $35.34 \leq$ Marks < 51.49

F = $19.19 \leq$ Marks < 35.34

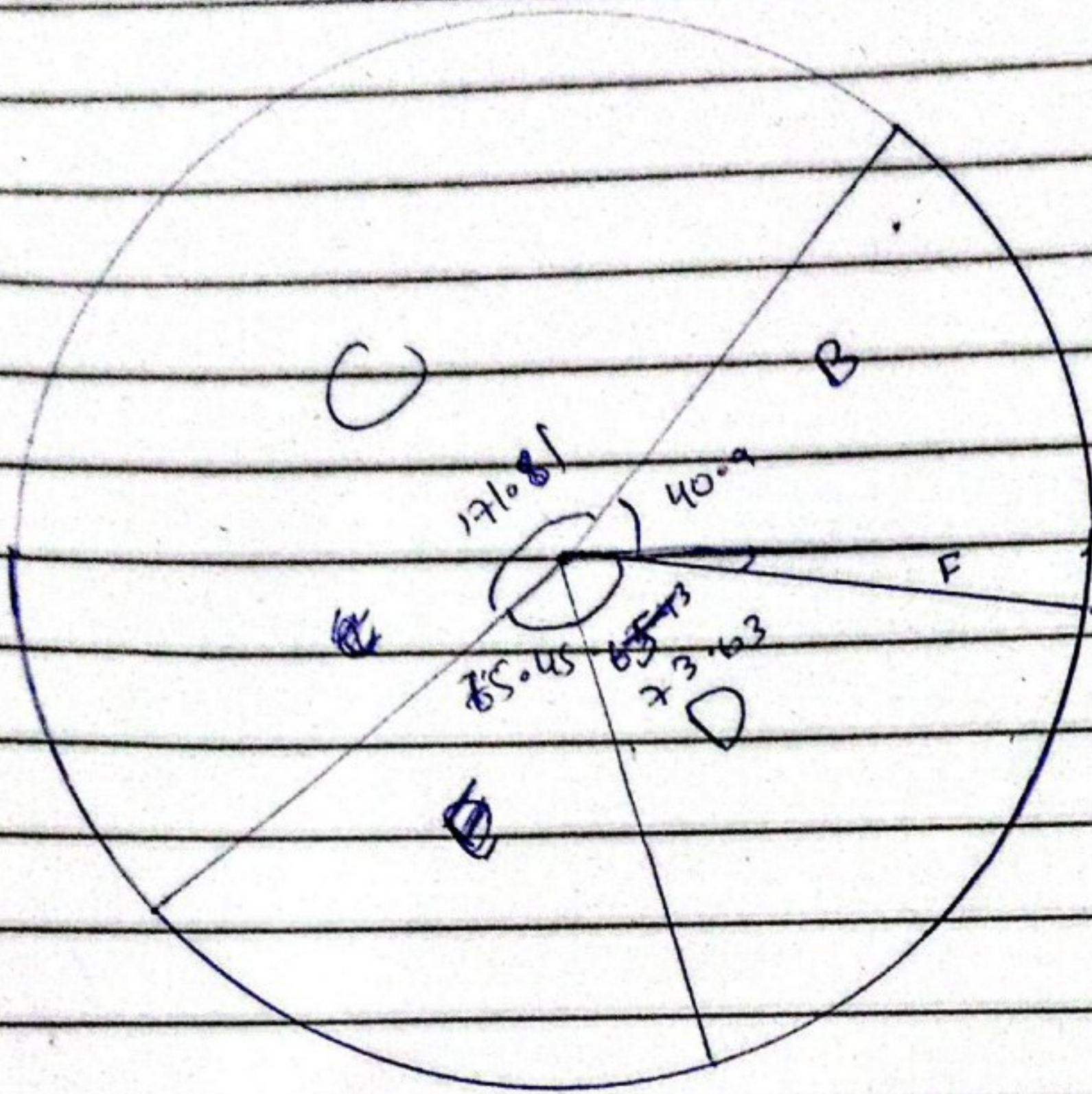
Grade	Freq	θ
A	0	0
B	5	40.9
C	21	171.81
D	9	73.63
E	8	65.45
F	1	8.18

(c) Bar Chart



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Piechart



Upon observing the graph we can say that most of the students had C grade and no student has grade A.